

MM: Merging Models via Important Neurons

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Abstract

The remarkable progress in deep learning has catalyzed the development and open-sourcing of a vast ecosystem of specialized models, each excelling in distinct tasks or domains. Despite their individual effectiveness, deploying a single fine-tuned model often proves insufficient to accommodate the heterogeneous and dynamic requirements of real-world applications. In response, *model merging*—the practice of integrating multiple pre-trained or fine-tuned models into a unified architecture without additional training—has emerged as a compelling paradigm for achieving both efficiency and performance scalability. Nevertheless, current model merging strategies frequently suffer from significant performance degradation on domain-specific benchmarks relative to their constituent source models. This limitation stems primarily from unresolved parameter conflicts and cross-task interference that arise when naively averaging or interpolating model weights across heterogeneous objectives.

To overcome these fundamental challenges, we introduce **MM**, a novel router-based framework that rethinks model merging through the lens of *neuron-level selectivity* and *structured integration*. At its core, MM operates under the principle that only a sparse subset of neurons in each expert model carries task-critical information, while the remainder may introduce redundancy or conflict during fusion. Our method first applies structured pruning to each source model to eliminate non-essential neurons, thereby distilling a compact and high-signal expert representation. Subsequently, these refined experts are merged in a dynamic, hierarchical fashion guided by a trainable routing mechanism that adaptively selects and combines the most informative neurons based on input context and task semantics.

We conduct comprehensive evaluations across a broad spectrum of benchmarks spanning both Computer Vision (CV) and Natural Language Processing (NLP), including domain-specialized tasks such as medical image segmentation, low-resource machine translation, and instruction-following dialogue. Empirical results demonstrate that MM consistently surpasses not only conventional merging baselines (e.g., TIES-Merging, Fisher-weighted averaging) but also the original fine-tuned source models on their respective in-domain objectives. Crucially, MM simultaneously preserves—or even enhances—generalization capabilities on out-of-domain data, attesting to its ability to harmonize specialization with robustness. These findings establish MM as a principled, scalable, and effective

approach to building versatile foundation models through intelligent, neuron-aware model composition.

CCS Concepts

• **Computing methodologies** → **Neural networks; Multi-task learning**; *Natural language processing; Computer vision.*

Keywords

Model Merging, Parameter Conflicts, Task Interference, Multi-task Learning, Deep Learning, Router Mechanism

ACM Reference Format:

Anonymous Author(s). 2026. MM: Merging Models via Important Neurons. In *Proceedings of Proceedings of the ACM International Conference on Multimedia (ACM Multimedia '26)*. ACM, New York, NY, USA, 10 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Large language models (LLMs) [2, 47] have demonstrated unprecedented capabilities across a wide range of natural language processing (NLP) tasks, including code generation [14, 40], mathematical reasoning [1, 29], and multilingual understanding [31]. With billions of parameters and extensive pretraining, modern LLMs exhibit strong generalization and adaptability to diverse downstream applications [5, 9, 17, 23, 28, 36, 42–46, 49]. As the ecosystem of open-source LLMs continues to expand [34], each model often excels in specific domains or tasks, raising a critical question: *how can we effectively harness the complementary strengths of multiple specialized models within a unified framework?*

To address this challenge, several paradigms for multi-model collaboration have been proposed, including model ensembling, cooperative inference, and model merging [11, 19, 21]. Among these, *model merging*—which integrates multiple models into a single parameterized architecture through weight-level operations—offers a particularly appealing trade-off between performance, efficiency, and deployability. Unlike ensembling, which incurs significant computational overhead, merged models enable fast, memory-efficient inference without requiring access to multiple checkpoints at test time [26, 38]. Despite its promise, model merging remains an underexplored yet high-impact direction, primarily due to challenges such as parameter misalignment, task interference, and loss of specialization during integration.

Existing model merging approaches can be broadly categorized into four families. (1) *Weight-averaging methods*, such as Fisher-Merging [30] and RegMean [20], improve naive averaging by weighting parameters using Fisher information matrices or activation statistics. (2) *Task vector-based methods* operate in the tangent space of a pre-trained base model, merging task-specific updates rather than full parameters; notable examples include Task Arithmetic [32, 33], TIES-Merging [51], and AdaMerging [53], where the former two mitigate sign and magnitude conflicts, while the latter adapts merging coefficients per layer. (3) *Preprocessing-based strategies*, exemplified by DARE [55], prune and rescale task vectors to reduce

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ACM Multimedia '26, Anonymous

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ACM ISBN 978-1-4503-XXXX-X/2018/06
<https://doi.org/XXXXXXX.XXXXXXX>

interference before merging. (4) *Router-based approaches*, such as Twin-Merging [55] and Free-Merging [50], introduce dynamic routing mechanisms to activate task-specific subnetworks, enabling input-adaptive model behavior.

While these methods have advanced the field, they often fail to fully preserve in-domain performance or generalize robustly out-of-domain—largely due to indiscriminate parameter fusion that overlooks neuron-level functional importance. In this work, we propose **MM** (*Merging via Important Neurons*), a router-guided framework that selectively merges only the most salient neurons across experts, thereby minimizing destructive interference while maximizing retained expertise.

To address the challenges described above, we propose MM, a framework specifically designed to mitigate parameter conflicts and task interference in model merging. The method comprises three key stages, following the initial preparation step in which multiple expert models are obtained via fine-tuning. In the first stage (**Expert Enhancement**), we identify neurons that are most critical to each expert’s specialized task, retaining these while pruning less important neurons [13]. As indicated in Table 2, this procedure not only strengthens the specialization of each expert but also reduces parameter conflicts, which primarily occur among overlapping neurons. Consequently, the smaller the subset of retained neurons across experts, the lower the potential for conflict during merging. The second stage (**Router Training**) involves constructing a multi-layer perceptron (MLP) router using data sampled from the training sets of the fine-tuned expert models. During the merging process, the router assigns input-dependent weights to each expert. In scenarios involving a large number of experts, only the top- k experts with the highest router scores are selected to participate in the merging process, ensuring efficiency and relevance. Finally, in the third stage (**Dynamic Layer-wise Merging**), each expert’s layers are partitioned into important-neuron and less-important-neuron layers. These layers are merged using distinct strategies according to the input. By limiting the merging to the most informative parameters, this approach not only accelerates the merging process but also further mitigates parameter conflicts, preserving the integrity of the experts’ specialized knowledge.

We empirically demonstrate the effectiveness of MM, as summarized in Figure 1b. First, we merge five Qwen2.5-0.5B-Instruct [?] models to validate its performance in the NLP domain. Next, we merge ten ViT-Base-Patch16-224 models to confirm its efficacy in computer vision tasks. To assess scalability, we merge five Qwen2.5-7B-Instruct [?] models, illustrating that the approach can be applied to larger models. Furthermore, by merging a classification model with a mathematical reasoning model, we show that MM supports cross-domain and cross-task integration. Finally, evaluation on the out-of-domain MMLU [16] benchmark demonstrates that the merged models exhibit strong generalization and robustness.

Our contributions can be summarized as follows:

- **Novel Framework:** We propose MM, a novel neuron-level model merging framework that integrates multiple models into a single unified model.
- **Superior Performance:** MM is both simple and effective, as demonstrated through extensive empirical validation on CV

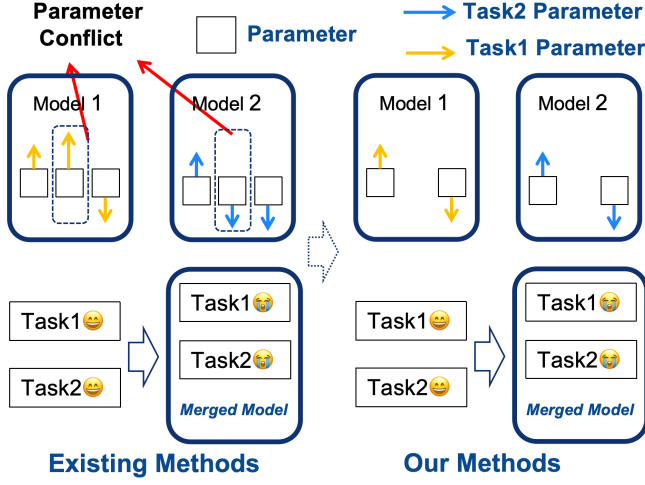
[6, 7, 35, 54] and NLP tasks, as well as on larger-scale models, while also exhibiting strong robustness on out-of-domain tasks.

- **Resolving Conflicts:** We fully resolve challenges of parameter conflicts and task interference in model merging, achieving performance that not only surpasses the original models on in-domain tasks but also preserves the generalization ability of pretrained models on out-of-domain tasks.

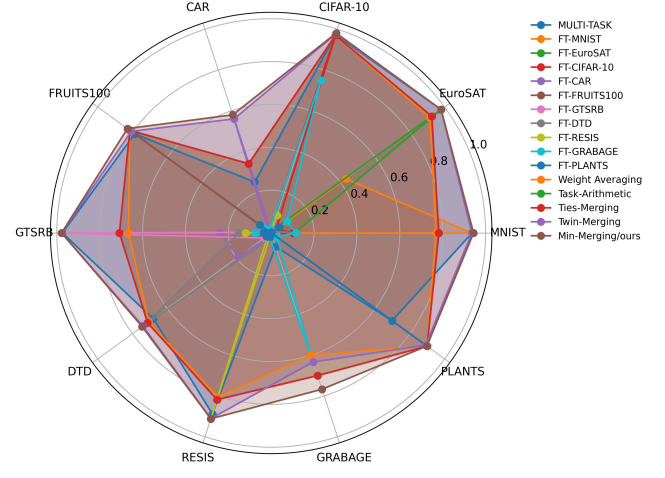
2 Related Work

Early approaches such as Fisher-Merging and RegMean apply simple weight averaging techniques but require access to extra data and computational resources. Other lines of work attempt to interpolate between models within a shared low-loss basin, guided by the theory of linear mode connectivity (LMC) [8, 10, 12]. Techniques like *weight matching* and *optimal transport* have been proposed to align parameters across networks for improved interpolation. However, recent studies suggest that LMC assumptions may not always hold in the context of fine-tuned models [18, 27, 37, 39]. Task-Arithmetic extends simple averaging to more flexible arithmetic operations in parameter space to better control model behavior, yet interference among multiple models remains a major challenge. To mitigate this, methods such as Ties-Merging, AdaMerging, and DARE [55] have been proposed to reduce task conflicts by resolving parameter redundancy, learning optimal merging coefficients, and reducing weight density. Twin-Merging [55] introduces a modular knowledge composition strategy that dynamically merges knowledge modules based on relevance. Some approaches assume access to task identity at test time and leverage task-specific cues to improve merging performance. However, such assumptions are often impractical in real-world scenarios, where task distribution is unknown or unpredictable. Moreover, none of the existing methods [11] completely resolve parameter contradictions and task interference. In contrast, our method addresses these issues through two key mechanisms. First, we alleviate parameter conflict by eliminating task-irrelevant neurons, thereby reducing redundant or contradictory parameters during merging. Second, we update the model’s internal communication mechanism to enable efficient expert collaboration across tasks, effectively mitigating task interference. These improvements collectively allow our approach to achieve strong generalization and scalability across diverse domains and task types.

Early methods such as Fisher-Merging and RegMean employ simple weight averaging but rely on additional data and computational resources. Another line of work interpolates models within a shared low-loss basin, guided by the theory of linear mode connectivity (LMC) [8, 10, 12], and improves interpolation via techniques like weight matching or optimal transport. However, LMC assumptions do not always hold in the context of fine-tuned models [18, 27, 37, 39]. Task-Arithmetic extends simple averaging to more flexible operations in parameter space to better control model behavior, yet interference among multiple models remains a major challenge. To mitigate conflicts, methods such as Ties-Merging, AdaMerging, and DARE [55] reduce task interference by eliminating redundant parameters, learning optimal merging coefficients, and lowering weight density, while Twin-Merging [55] introduces a modular knowledge composition strategy to dynamically merge



(a) The impact of parameter conflicts and task interference on model merging performance



(b) Comparison accuracy (%) with existing methods demonstrating the superior effectiveness of our approach

Figure 1: Illustration of Parameter Conflicts and Performance of the Proposed Method

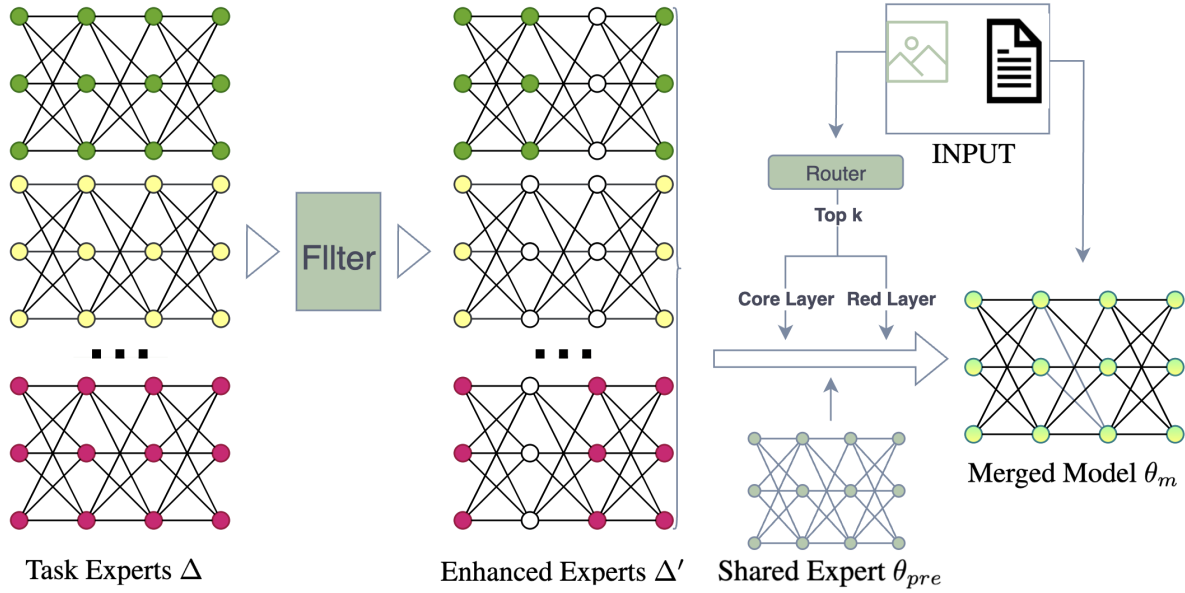


Figure 2: Overview of the proposed method, illustrating its main workflow.

relevant knowledge modules. Despite these advances, parameter conflicts and task interference remain largely unresolved. To address this, we propose a neuron-level merging approach that mitigates parameter contradictions and reduces task interference at their source, offering a more complete solution for multi-task model integration.

3 Method

This chapter introduces the overall workflow of our proposed method, which begins with the preliminary work and subsequently

proceeds through three core stages, as illustrated in Figure 2. In Section 3.1, we enhance the expert models by identifying and retaining their most important neurons. In Section 3.2, we carry out targeted router training to ensure that, during inference, the merged model can accurately route inputs to the appropriate expert clusters. Finally, in Section 3.3, we propose a dynamic layer-wise merging strategy to fully exploit the strengthened experts and the router network, thereby enabling more effective integration across expert clusters.

Task denotes. Given N tasks $[T_1, \dots, T_N]$, the goal of model merging is to obtain a single model suitable for all tasks by using the models $[\theta_1, \dots, \theta_N]$ fine-tuned from the same pretrained model θ_{pre} . Existing methods focus on merging these models into a unified model θ_m . It is important to note that we adopt LoRA as an efficient fine-tuning method, which is more compatible with our approach. Compared with full-parameter fine-tuning, LoRA can reduce memory consumption during inference and improve inference speed.

3.1 Expert Enhancement

Inspired by the task arithmetic approach, we can write

$$\theta_m = \theta_{pre} + \sum \alpha_i (\theta_i - \theta_{pre}), \quad (1)$$

where $\theta_i - \theta_{pre}$ represents the expert knowledge Δ_i for each domain. By **condensing** the parameters corresponding to each expert, it is possible for the expert to achieve higher specialization in its domain using fewer parameters. This not only reduces parameter conflicts during the merging process but also effectively raises the pre-merging starting point for each expert, thereby enhancing the final performance of the merged model.

Inspired by pruning strategies, we enhance the specialization of each expert model by removing redundant layers. In our approach, pruning is performed **at the layer level**, which not only concentrates the model's capacity on task-relevant layers but also naturally couples with the dynamic layer-wise merging strategy introduced in Subsection 3.3.

To empirically validate our approach, we conducted preliminary experiments on LoRA-based models trained for the RTE, MNLI, MRPC, and QNLI tasks [48]. The corresponding expert models were further specialized by automatically pruning LoRA parameters in selected layers. As illustrated in Figure 3, the RTE expert achieved improved performance after pruning layers 13-21, the QNLI expert after layers 14-16, the MNLI expert after layers 2-5, and the MRPC expert after layers 16-21. Further details are provided in Table ?? ?? . Notably, the entire pruning process was fully automated.

Formally, for the i -th expert Δ_i with L layers, we partition its layers into core and redundant sets:

$$\mathcal{L}_i = \mathcal{L}_i^{\text{core}} \cup \mathcal{L}_i^{\text{redundant}}, \quad \mathcal{L}_i^{\text{core}} \cap \mathcal{L}_i^{\text{redundant}} = \emptyset, \quad (2)$$

where $\mathcal{L}_i^{\text{core}}$ denotes the retained core layers, and $\mathcal{L}_i^{\text{redundant}}$ denotes the pruned redundant layers. After pruning, the expert model is reduced to

$$\Delta'_i = \mathcal{L}_i^{\text{core}}, \quad (3)$$

From an information-theoretic perspective, each layer can be regarded as an information carrier for task T_i . The core layers $\mathcal{L}_i^{\text{core}}$ contain the most task-relevant signal, while the redundant layers $\mathcal{L}_i^{\text{redundant}}$ mainly contain noise. We can quantify this using the **signal-to-noise ratio (SNR)** [22]:

$$\text{SNR}_i = \frac{\sum_{l \in \mathcal{L}_i^{\text{core}}} I(\Delta_i^{(l)}; T_i)}{\sum_{l \in \mathcal{L}_i^{\text{redundant}}} H(\Delta_i^{(l)})}, \quad (4)$$

where $I(\Delta_i^{(l)}; T_i)$ denotes the mutual information between the l -th layer and task T_i , and $H(W_i^{(l)})$ denotes the entropy of the redundant layer parameters. By pruning redundant layers, the denominator

decreases while the numerator is maintained or concentrated, resulting in

$$\text{SNR}_i \gg 1. \quad (5)$$

A higher SNR indicates that the expert's effective information is concentrated in the most relevant layers, enhancing specialization and reducing parameter conflicts during merging.

3.2 Router Training

Inspired by Twin-Merging, we introduce a **Router** module to dynamically select the most relevant subset of experts. When the number of experts is large, only the top- k experts most relevant to the input are activated, improving computational efficiency and reducing interference from irrelevant experts.

The Router is essentially a lightweight multi-layer perceptron (MLP), trained as a multi-class classification task. Let the input be x and the task label be $y \in \{1, \dots, N\}$, the Router outputs a probability distribution over experts:

$$p(x) = \text{softmax}(g_\theta(x)) \in \mathbb{R}^N, \quad (6)$$

where g_θ is the Router forward function and θ denotes its parameters. The training objective is to minimize the cross-entropy loss:

$$\mathcal{L}_{\text{router}} = - \sum_{i=1}^N \mathbf{1}_{[y=i]} \log p_i(x). \quad (7)$$

During inference, the top- k experts are selected:

$$\mathcal{E}_{\text{topk}}(x) = \text{TopK}(p(x), k), \quad (8)$$

And the expert coefficients produced by the Router, $\mathcal{E}_{\text{topk}}(x)$, are subsequently used in Subsection 3.3 to guide the dynamic layer-wise merging process.

3.3 Dynamic Layer-wise Merging

In this subsection, we present the final stage of our method: dynamic layer-wise merging. The term "dynamic" refers to the input-dependent, real-time adaptation of the merging process. Specifically, the Router module selects the top- k most relevant experts for each input, ensuring that inference fully leverages the most pertinent expert knowledge while incurring minimal additional computational overhead.

The term "hierarchical" refers to differentiating between core and redundant layers during merging. Core layers preserve the primary domain expertise, whereas redundant layers incorporate cross-domain information, reducing computational cost and mitigating parameter conflicts. For instance, in the context of solving mathematical problems, mathematical reasoning represents the primary domain expertise (core layers), while language comprehension constitutes cross-domain knowledge (redundant layers) that facilitates understanding problem statements.

Formally, let the Router output for top- k experts be $\mathcal{E}_{\text{topk}}$. The experts are then ranked according to their relevance:

$$(\ell, \mathcal{E}) = \text{argsort}_{\downarrow}(\mathcal{E}_{\text{topk}}) \quad (9)$$

The parameters of the merged model are given by:

$$\theta_m = \theta_{pre} + \left(\mathcal{E}_1 \mathcal{L}_{\ell_1}^{\text{core}} \cup \sum_{i=2}^k \mathcal{E}_i \mathcal{L}_{\ell_i}^{\text{redundant}} \right) \quad (10)$$

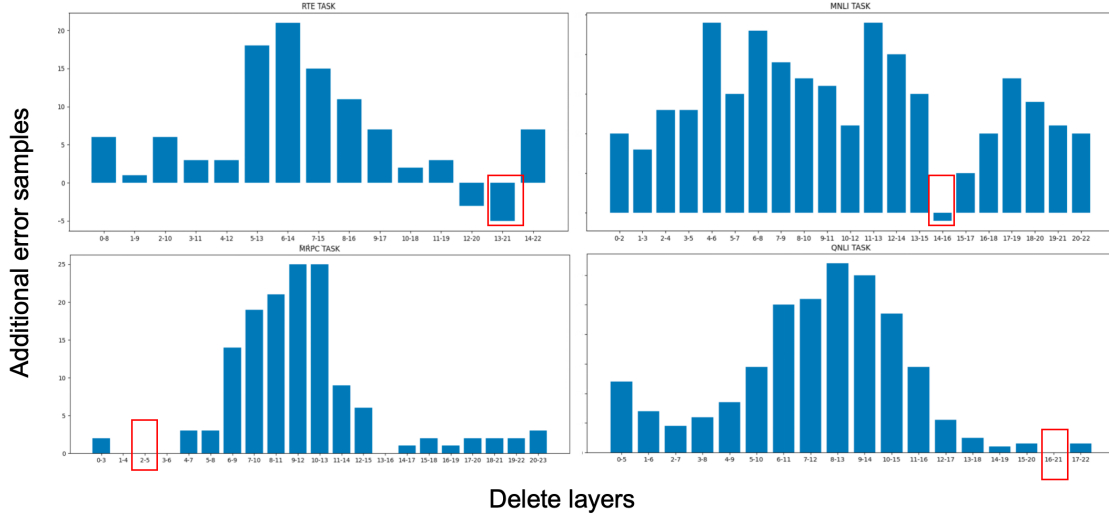


Figure 3: Performance of expert models after pruning LoRA parameters in different layers. The horizontal axis denotes the pruned layers, and the vertical axis represents the corresponding performance drop.

where $\mathcal{L}_{\ell_1}^{\text{core}}$ denotes the core layers of the most relevant expert, $\mathcal{L}_{\ell_1}^{\text{redundant}}$ denotes the redundant layers of the other experts, and $\mathcal{E}_i$ represents the corresponding weights assigned by the Router.

From a signal perspective, the output of each expert e can be decomposed as:

$$f_e(\mathcal{L}, x) = s_e(\mathcal{L}, x) + n_e(\mathcal{L}, x) \quad (11)$$

where $s_e(\mathcal{L}, x)$ denotes the task-relevant signal, and $n_e(\mathcal{L}, x)$ denotes noise or irrelevant components.

Consequently, the effective signal-to-noise ratio (SNR) of the merged model can be approximated as:

$$\text{SNR}_{\text{merging}}(x) \approx \frac{\sum_{e \in \mathcal{E}_{\text{topk}_1}} \alpha_e^2 \mathbb{E}[|s_e(\mathcal{L}_{\text{core}}, x)|^2] + \sum_{e \in \mathcal{E}_{\text{topk}_2 - \text{topk}}} \beta_e^2 \mathbb{E}[|s_e(\mathcal{L}_{\text{redundant}}, x)|^2]}{\sum_{e \in \mathcal{E}_{\text{topk}_1}} \alpha_e^2 \mathbb{E}[|n_e(\mathcal{L}_{\text{core}}, x)|^2] + \sum_{e \in \mathcal{E}_{\text{topk}_2 - \text{topk}}} \beta_e^2 \mathbb{E}[|n_e(\mathcal{L}_{\text{redundant}}, x)|^2]} \quad (12)$$

This framework ensures that the core layers primarily contribute task-specific knowledge, while the redundant layers provide complementary cross-domain information, with noise being effectively attenuated. As a result, the overall performance of the merged model is enhanced.

4 Experiment

In this section, we comprehensively evaluate the performance of our proposed MM method under various experimental settings, including cross-task, cross-domain, cross-training configurations, and domain-shift scenarios. We compare our approach against three representative model merging baselines: **Weight Averaging**, **Task Arithmetic**, **Ties-Merging** and **Twin-Merging**. Section 4.1 presents the performance of our proposed method on both NLP and CV benchmarks. Section 4.2 evaluates its scalability to a larger number of models and examines cross-domain and cross-task generalization capabilities. Section 4.3 conducts ablation studies to quantify the contribution of each component to the overall performance.

4.1 Comparative Evaluation

Setup. We conduct experiments on five NLP datasets: RTE, MNLI, QNLI, QQP, MRPC [48], and ten CV datasets: MNIST [25], EuroSAT [15], CIFAR-10 [24], CarBrands50, Fruits100, GTSRB [41], DTD [4], RESISC [3], GRABAGE, PLANTS.

For NLP experiments, we apply our proposed method and all baselines to the Qwen2.5-0.5B-Instruct model [52?], whereas for CV tasks, the same methods are evaluated using the pretrained ViT-Base-Patch16-224 architecture. Unless stated otherwise, all input images are uniformly resized to 224×224 resolution for both training and inference. Across both domains, we adopt consistent evaluation protocols: throughput is measured on a single NVIDIA A100 GPU with a batch size of 32 under FP32 precision, classification accuracy is used as the primary evaluation metric, and computational cost is assessed in terms of memory usage and inference time. A more detailed overview of the datasets and implementation details is provided in Appendix ??.

Results. As shown in Table 1 2, both on NLP and CV tasks, our method not only outperforms existing model merging baselines but, quite unexpectedly, also surpasses task-specific fine-tuned models in single-task performance with minimal cost of storage and time. This finding strongly supports the effectiveness of our approach in resolving common issues in model merging, such as parameter conflicts and task interference. In contrast to prior methods that often fail to scale due to these challenges, our approach exhibits superior robustness and scalability.

4.2 Comparative Analysis

Large-Model Scalability and Out-of-Domain Generalizability. Merging larger-scale models is an important issue that has received increasing attention. To evaluate the scalability of our approach, we conducted experiments with Qwen2.5-7B-Instruct under the same settings as Qwen2.5-0.5B-Instruct. As shown in Table ??, our

Table 1: Comparison of task-specific fine-tuned models, merge baselines, and our method across GLUE tasks.

MODEL	RTE	MNLI	QNLI	QQP	MRPC	AVG	VRAM	TIME
MULTI-TASK	77.4%	81.1%	83.0%	78.0%	76.2%	79.1%	2010M	184s
FT-RTE	77.3%	52.8%	56.0%	61.0%	58.9%	61.2%	2010M	195s
FT-MNLI	71.5%	82.0%	33.4%	62.0%	63.3%	62.4%	2010M	195s
FT-QNLI	62.5%	46.8%	84.0%	65.0%	65.7%	64.8%	2010M	195s
FT-QQP	64.3%	43.2%	64.4%	84.8%	70.4%	65.4%	2010M	195s
FT-MRPC	49.5%	36.8%	56.0%	65.8%	85.3%	58.7%	2010M	195s
Weight Averaging	68.2%	39.6%	67.8%	64.0%	57.1%	59.4%	2010M	195s
Task-Arithmetic	66.8%	65.6%	59.2%	71.6%	74.0%	66.4%	2010M	195s
Ties-Merging	66.4%	65.2%	59.6%	70.4%	67.9%	65.9%	2010M	195s
Twin-Merging	76.9%	81.8%	83.6%	84.8%	85.0%	82.4%	2010M+(N-1)*34.2M	275s
MM/ours	79.1%	82.4%	84.4%	85.2%	85.3%	83.3%	2010M+(N-1)*20M	235s

method again achieves remarkable performance, surpassing fine-tuned baselines, indicating that its effectiveness is not constrained by model size. Out-of-domain (OOD) generalization has long been a major challenge for model merging, since the process primarily integrates a limited set of expert capabilities. As a result, it is natural for merged models to experience performance drops when facing capabilities beyond those experts. However, as reported in Table 4 (with detailed results in Table ??), when evaluated on MMLU, the merged Qwen2.5-7B-Instruct models trained on RTE, MNLI, QNLI, QQP, and MRPC outperform both Weight Averaging and Task-Arithmetic, and achieve performance nearly on par with general-purpose models. This surprising result provides strong evidence that MM retains robust OOD generalization.

Scalability to Large-Scale Tasks. As illustrated in the left panel of Figure 4, the curve shows how the average regularization metric evolves with the number of tasks. Unlike prior methods whose performance declines as the number of models increases, our approach maintains stable regularization effectiveness and continues to outperform fine-tuned models. In the right panel of Figure 4, our method exhibits lower memory consumption compared to Twin-Merging. Moreover, as reported in Tables 1 and 2, our inference time is consistently shorter than that of Twin-Merging. These results demonstrate that our method achieves superior performance at significantly lower memory and time costs, enabling the integration of a substantially larger number of models while surpassing the theoretical upper bound of fine-tuned baselines. More details show in Appendix ??.

Task Diversity and Span. As shown in Tables 1 2, our method demonstrates robust performance across both computer vision (CV) and natural language processing (NLP) tasks, covering a diverse set of classification scenarios such as 2 (GARBAGE), 3 (RTE), 10 (EuroSAT), 30 (PLANTS), 43 (GTSRB), 45 (RESISC), 47 (DTD), 50 (CarBrands50), and 100 (Fruits100). These tasks span a broad range of semantic domains, including digits, remote sensing, general objects, automobiles, fruits, plants, waste, traffic signs and various other domains, highlighting the method’s ability to generalize

across distinct knowledge areas. Furthermore, as illustrated in Table 3, our approach supports the integration of both generative and discriminative tasks within a unified framework, marking a pioneering effort in this direction. Notably, it even outperforms fine-tuned models that are often regarded as the theoretical upper bound. These results collectively underscore the exceptional scalability of our method across modalities, knowledge domains, and task types.

4.3 Ablation Studies

To assess the effectiveness of each component in our method, we perform ablation studies on five NLP and five CV tasks. Specifically, we examine the impact of three core modules: Expert Enhancement, Hierarchical Merging and Router.

Expert Enhancement Analysis. Disabling expert enhancement means that important neurons are no longer preserved to strengthen experts, and consequently, the hierarchical strategy built upon it cannot be applied. As shown in Table 5, performance on NLP tasks drops from 83.3% to 82.2%, falling below the fine-tuned baseline of 82.7%. While the absolute performance drop may appear modest, its significance is substantial—expert enhancement enables the merged model to surpass the performance of fine-tuned models, which were previously regarded as the theoretical upper bound in model merging. This improvement stems from the reduction of parameter conflicts during merging, as well as the elevation of the merging starting point.

Hierarchical Merging Analysis. Disabling hierarchical merging means treating all layers as equally important and merging them using a uniform strategy. As shown in Table 5, accuracy on NLP tasks slightly decreases from 83.3% to 82.8%, while performance on CV tasks drops from 86.6% to 86.1%. Despite this limited decline, our method still surpasses fine-tuned baselines (82.7% for NLP and 85.3% for CV), indicating that expert enhancement alone can yield substantial benefits. However, hierarchical merging provides an additional performance boost beyond the fine-tuned upper bound.

Table 2: Comparison of task-specific fine-tuned models, merge baselines, and our method across CV tasks.

MODEL	MNIST	EuroSAT	CIFAR-10	CAR	FRUITS100	GTSRB	DTD	RESIS	GRABAGE	PLANTS	AVG	VRAM
TIME												
MULTI-TASK	93.5%	98.1%	97.1%	46.0%	81.6%	98.2%	69.7%	91.2%	66.7%	74.1%	75.6%	805M
158s												
FT-MNIST	93.6%	4.3%	0.1%	0.0%	0.1%	0.6%	0.4%	0.2%	0.0%	0.1%	9.9%	805M
158s												
FT-EuroSAT	10.9%	98.0%	1.1%	0.0%	0.4%	4.0%	1.2%	2.7%	0.0%	1.1%	11.9%	805M
158s												
FT-CIFAR-10	10.2%	2.8%	97.8%	2.0%	2.9%	4.1%	0.7%	2.9%	0.0%	0.6%	12.4%	805M
158s												
FT-CAR	0.0%	1.0%	0.1%	56.0%	1.5%	2.4%	1.9%	1.6%	0.0%	0.6%	6.5%	805M
158s												
FT-FRUIT100	1.1%	1.9%	0.1%	0.0%	80.7%	0.4%	0.0%	0.2%	0.0%	1.9%	8.6%	805M
158s												
FT-GTSRB	2.3%	0.1%	0.6%	0.0%	0.5%	97.8%	0.4%	0.8%	0.0%	0.1%	10.3%	805M
158s												
FT-DTD	11.8%	1.2%	1.8%	1.0%	0.1%	1.5%	73.8%	3.2%	0.0%	2.6%	9.7%	805M
158s												
FT-RESIS	0.0%	0.0%	8.7%	1.0%	0.4%	1.2%	1.9%	91.1%	0.0%	0.0%	10.4%	805M
158s												
FT-GRABAGE	11.6%	9.2%	7.5%	2.0%	5.0%	7.3%	2.2%	2.2%	63.3%	3.3%	11.4%	805M
158s												
FT-PLANTS	0.2%	4.6%	0.4%	0.0%	6.5%	3.5%	2.8%	1.9%	0.0%	89.1%	10.9%	805M
158s												
Weight Averaging	76.9%	91.6%	96.8%	34.0%	81.6%	66.6%	70.6%	80.6%	60.0%	89.8%	74.8%	805M
158s												
Task-Arithmetic	78.2%	92.6%	97.0%	34.0%	81.5%	70.8%	71.3%	81.8%	70.0%	89.9%	76.7%	805M
158s												
Ties-Merging	78.2%	92.6%	97.0%	34.0%	81.5%	70.8%	71.3%	81.8%	70.0%	89.9%	76.7%	805M
158s												
Twin-Merging	93.6%	98.0%	97.8%	56.0%	80.7%	97.8%	74.0%	91.1%	63.3%	89.1%	84.1%	805M+(N-1)*7M
223s												
MM/ours	94.4%	98.0%	98.1%	58.0%	82.9%	97.8%	74.4%	91.2%	76.7%	89.8%	86.6%	805M+(N-1)*5M
179s												

Table 3: Performance Across Different Task Types

Method	Classification	Generation
Finetune	77.3%	49.5%
MM/ours	79.1%	51.5%

Table 4: Average Performance on Out-of-Domain MMLU Benchmark Tasks

MMLU Task	Qwen2.5-7B	Weight Avg.	Task-Arith.	MM(Ours)
Avg	67.9%	67.3%	63.6%	67.5%

This is achieved by partitioning the model into core and redundant layers, and applying differentiated merging strategies: core

Table 5: Ablation study of the proposed MM method, evaluating the contribution of filtering, hierarchical routing performance across NLP and Vision tasks.

Method	NLP(Avg.)	CV(Avg.)
Finetune	82.7%	85.3%
w/o Filtering and Hierarchical	82.2%	84.3%
w/o Hierarchical	82.8%	86.1%
w/o Router	64.2%	78.0%
MM	83.3%	86.6%

layers preserve domain-specific knowledge, while redundant layers integrate cross-domain information. For example, solving a math

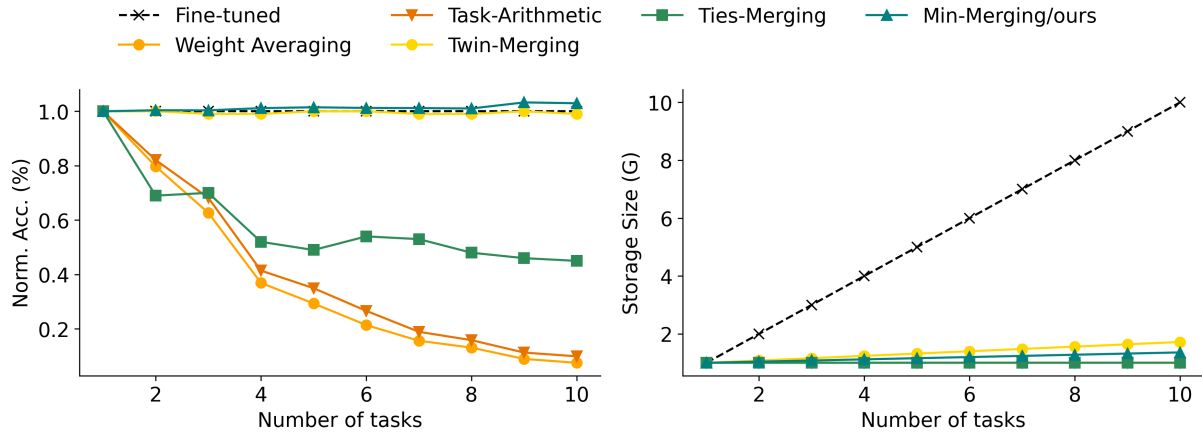


Figure 4: Scalability analysis of model accuracy and storage footprint as the number of tasks increases.

problem requires not only mathematical reasoning (domain-specific knowledge) but also language comprehension to interpret the question (cross-domain knowledge). Our hierarchical approach ensures that both types of knowledge are efficiently integrated into the merged model.

Router Analysis. The **Router** module forms the foundation of our dynamic merging. As shown in Table 3, performance on NLP tasks drops significantly from 83.3% to 64.2%, while accuracy on CV tasks decreases from 86.6% to 78.0%. This substantial performance decline clearly demonstrates the critical role of the Router module. By enabling efficient internal communication, the Router dynamically activates the top- k most important expert modules for each input. The most important expert takes the lead in task processing, while the remaining $k - 1$ experts provide supplementary support through collaboration. This strategy not only significantly improves performance but also offers strong scalability: even when merging a larger pool of experts, the Router ensures that only a small subset of highly important experts is activated, enabling efficient collaboration across a large-scale expert pool.

5 Conclusion

In this work, we introduce **MM**, a novel model merging framework specifically designed to mitigate parameter conflicts that commonly arise when integrating multiple specialized models. Our approach proceeds in two stages: first, each expert model undergoes enhancement through structured pruning to alleviate internal redundancies and reduce inter-model contradictions; second, a dynamic, layer-wise merging strategy guided by task-aware routing selectively fuses only the most informative neurons. Comprehensive experiments across diverse computer vision and natural language processing benchmarks demonstrate that **MM** not only consistently surpasses the performance of individually fine-tuned source models on their respective in-domain tasks but also achieves this with minimal computational overhead—marking the first merging method to do so. Furthermore, **MM** preserves strong out-of-domain generalization capabilities comparable to those of general-purpose pretrained models. Additional analyses confirm its scalability to

larger architectures and an increasing number of experts, as well as its robustness in challenging cross-task and cross-domain merging scenarios.

6 Limitations and Future

Although our proposed **MM** method has achieved promising results, it is important to acknowledge certain limitations.

One of the fundamental challenges in model merging lies in integrating models with heterogeneous architectures. While **MM** performs effectively when merging models from similar architectural families or pretraining strategies, its generalizability and performance on models with significantly different backbones or initialization methods remain to be explored.

Looking forward, we identify several promising directions for future work:

- **Scalability Validation:** Although **MM** is theoretically capable of merging thousands of models, due to limitations in computational resources and training time, we have not yet demonstrated this scalability empirically. In future work, we plan to conduct large-scale experiments to validate this capability.
- **Integration with MoE Architectures:** Mixture-of-Experts (MoE) [56–58] architectures have emerged as a dominant paradigm in large-scale model design. While MoE models offer impressive inference efficiency, they are often resource-intensive and difficult to train. An interesting research direction is to explore whether **MM** can be used to pre-integrate expert modules into MoE architectures, thereby accelerating convergence and improving training stability.
- **Toward AGI-Level Integration:** Since **MM** effectively lifts previous limitations on the number of models that can be merged, it opens up the possibility of constructing a unified model capable of excelling across diverse domains. In the future, if a common architecture demonstrates strong performance across modalities such as speech, vision, nlp, and graph reasoning, **MM** could serve as a foundation to integrate all task-specific knowledge into one system—bringing us a step closer to the era of Artificial General Intelligence (AGI).

References

- [1] Zhangir Azerbayev, Hailey Schoelkopf, Keiran Paster, Marco Dos Santos, Stephen McAleer, Albert Q Jiang, Jia Deng, Stella Biderman, and Sean Welleck. 2023. Llemma: An open language model for mathematics. *arXiv preprint arXiv:2310.10631* (2023).
- [2] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. arXiv:2005.14165 [cs.CL] <https://arxiv.org/abs/2005.14165>
- [3] Gong Cheng, Junwei Han, and Xiaoqiang Lu. 2017. Remote sensing image scene classification: Benchmark and state of the art. *Proc. IEEE* 105, 10 (2017), 1865–1883.
- [4] Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andreea Vedaldi. 2014. Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 3606–3613.
- [5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 4171–4186.
- [6] Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. 2020. Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping. arXiv:2002.06305 [cs.CL] <https://arxiv.org/abs/2002.06305>
- [7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. arXiv:2010.11929 [cs.CV] <https://arxiv.org/abs/2010.11929>
- [8] Felix Draxler, Kambis Veschgini, Manfred Salmhofer, and Fred Hamprecht. 2018. Essentially no barriers in neural network energy landscape. In *International conference on machine learning*. PMLR, 1309–1318.
- [9] Chenghao Fan, Zhenyi Lu, Wei Wei, Jie Tian, Xiaoye Qu, Danyang Chen, and Yu Cheng. 2024. On Giant’s Shoulders: Effortless Weak to Strong by Dynamic Logits Fusion. *arXiv preprint arXiv:2406.15480* (2024).
- [10] Jonathan Frankle, Gintare Karolina Dziugaite, Daniel Roy, and Michael Carbin. 2020. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*. PMLR, 3259–3269.
- [11] Ruiji Fu, Zhengqi Pei, Jiefu Gong, Wei Song, Dechuan Teng, Wanxiang Che, Shijin Wang, Guoping Hu, and Ting Liu. 2018. Chinese grammatical error diagnosis using statistical and prior knowledge driven features with probabilistic ensemble enhancement. In *Proceedings of the 5th Workshop on Natural Language Processing Techniques for Educational Applications*. 52–59.
- [12] Timur Garipov, Pavel Izmilov, Dmitrii Podoprikin, Dmitry P Vetrov, and Andrew G Wilson. 2018. Loss surfaces, mode connectivity, and fast ensembling of dnns. *Advances in neural information processing systems* 31 (2018).
- [13] Amirata Ghorbani and James Y Zou. 2020. Neuron shapley: Discovering the responsible neurons. *Advances in neural information processing systems* 33 (2020), 5922–5932.
- [14] Daya Guo, Qihao Zhu, Dejian Yang, Zhenda Xie, Kai Dong, Wentao Zhang, Guanting Chen, Xiao Bi, Yu Wu, YK Li, et al. 2024. DeepSeek-Coder: When the Large Language Model Meets Programming—The Rise of Code Intelligence. *arXiv preprint arXiv:2401.14196* (2024).
- [15] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. 2019. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (2019).
- [16] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring Massive Multitask Language Understanding. *Proceedings of the International Conference on Learning Representations (ICLR)* (2021).
- [17] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. 2022. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556* (2022).
- [18] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. 2022. Lora: Low-rank adaptation of large language models. *ICLR* 1, 2 (2022), 3.
- [19] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. 2023. Editing Models with Task Arithmetic. arXiv:2212.04089 [cs.LG] <https://arxiv.org/abs/2212.04089>
- [20] Xisen Jin, Xiang Ren, Daniel Preotiuc-Pietro, and Pengxiang Cheng. 2022. Dataless knowledge fusion by merging weights of language models. *arXiv preprint arXiv:2212.09849* (2022).
- [21] Xisen Jin, Xiang Ren, Daniel Preotiuc-Pietro, and Pengxiang Cheng. 2025. Dataless Knowledge Fusion by Merging Weights of Language Models. arXiv:2212.09849 [cs.CL] <https://arxiv.org/abs/2212.09849>
- [22] Don H Johnson. 2006. Signal-to-noise ratio. *Scholarpedia* 1, 12 (2006), 2088.
- [23] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361* (2020).
- [24] Alex Krizhevsky. 2009. *Learning multiple layers of features from tiny images*. Technical Report.
- [25] Yann LeCun, Corinna Cortes, and CJ Burges. 2010. MNIST handwritten digit database. *ATT Labs [Online]*. Available: <http://yann.lecun.com/exdb/mnist> 2 (2010).
- [26] Lucas Liebenwein, Cenk Baykal, Brandon Carter, David Gifford, and Daniela Rus. 2021. Lost in pruning: The effects of pruning neural networks beyond test accuracy. *Proceedings of Machine Learning and Systems* 3 (2021), 93–138.
- [27] Haokun Liu, Derek Tam, Mohammed Muqeeth, Jay Mohta, Tenghao Huang, Mohit Bansal, and Colin A Raffel. 2022. Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. *Advances in Neural Information Processing Systems* 35 (2022), 1950–1965.
- [28] Zhenyi Lu, Jie Tian, Wei Wei, Xiaoye Qu, Yu Cheng, Danyang Chen, et al. 2024. Mitigating boundary ambiguity and inherent bias for text classification in the era of large language models. *arXiv preprint arXiv:2406.07001* (2024).
- [29] Haipeng Luo, Qingfeng Sun, Can Xu, Pu Zhao, Jianguang Lou, Chongyang Tao, Xiubo Geng, Qingwei Lin, Shifeng Chen, and Dongmei Zhang. 2023. Wizardmath: Empowering mathematical reasoning for large language models via reinforced evol-instruct. *arXiv preprint arXiv:2308.09583* (2023).
- [30] Michael Matena and Colin Raffel. [n. d.]. Merging models with fisher-weighted averaging, 2021. *arXiv preprint arXiv:2111.09832* ([n. d.]).
- [31] Taishi Nakamura, Mayank Mishra, Simone Tedeschi, Yekun Chai, Jason T Stillerman, Felix Friedrich, Prateek Yadav, Tanmay Laud, Vu Minh Chien, Terry Yue Zhuo, et al. 2024. Aurora-m: The first open source multilingual language model red-teamed according to the us executive order. *arXiv e-prints* (2024), arXiv:2404.04100.
- [32] Guillermo Ortiz-Jimenez, Alessandro Favero, and Pascal Frossard. 2023. Task arithmetic in the tangent space: Improved editing of pre-trained models. *Advances in Neural Information Processing Systems* 36 (2023), 66727–66754.
- [33] Guillermo Ortiz-Jimenez, Alessandro Favero, and Pascal Frossard. 2023. Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models. arXiv:2305.12827 [cs.LG] <https://arxiv.org/abs/2305.12827>
- [34] David Patterson, Joseph Gonzalez, Quoc Le, Chen Liang, Lluís-Miquel Munguia, Daniel Rothchild, David So, Maud Texier, and Jeff Dean. 2021. Carbon emissions and large neural network training. *arXiv preprint arXiv:2104.10350* (2021).
- [35] Sayak Paul and Pin-Yu Chen. 2021. Vision Transformers are Robust Learners. arXiv:2105.07581 [cs.CV] <https://arxiv.org/abs/2105.07581>
- [36] Zhengqi Pei, Zhewei Sun, and Yang Xu. 2019. Slang detection and identification. In *Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL)*. 881–889.
- [37] Zhengqi Pei and Shuhui Wang. 2023. Dynamics-inspired neuromorphic visual representation learning. In *International Conference on Machine Learning*. PMLR, 27521–27541.
- [38] Zhengqi Pei, Anran Zhang, Shuhui Wang, and Qingming Huang. 2024. Modeling language tokens as functionals of semantic fields. In *Forty-first International Conference on Machine Learning*.
- [39] Zhengqi Pei, Anran Zhang, Shuhui Wang, Xiangyang Ji, and Qingming Huang. 2024. Data-free neural representation compression with riemannian neural dynamics. In *Forty-first International Conference on Machine Learning*.
- [40] Baptiste Roziere, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi Adi, Jingyu Liu, Romain Sauvestre, Tal Remez, et al. 2023. Code llama: Open foundation models for code. *arXiv preprint arXiv:2308.12950* (2023).
- [41] Johannes Stalkamp, Marc Schlipf, Jan Salmen, and Christian Igel. 2011. The German traffic sign recognition benchmark: a multi-class classification competition. In *The 2011 international joint conference on neural networks*. IEEE, 1453–1460.
- [42] Zhaochen Su, Juntao Li, Jun Zhang, Tong Zhu, Xiaoye Qu, Pan Zhou, Yan Bowen, Yu Cheng, et al. 2024. Living in the Moment: Can Large Language Models Grasp Co-Temporal Reasoning? *arXiv preprint arXiv:2406.09072* (2024).
- [43] Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshuo Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. 2024. Conflictbank: A benchmark for evaluating the influence of knowledge conflicts in llm. *arXiv preprint arXiv:2408.12076* (2024).
- [44] Zhaochen Su, Jun Zhang, Tong Zhu, Xiaoye Qu, Juntao Li, Min Zhang, and Yu Cheng. 2024. Timo: Towards better temporal reasoning for language models. *arXiv preprint arXiv:2406.14192* (2024).
- [45] Xiaofei Sun, Linfeng Dong, Xiaoya Li, Zhen Wan, Shuhe Wang, Tianwei Zhang, Jiwei Li, Fei Cheng, Lingjuan Lyu, Fei Wu, et al. 2023. Pushing the limits of chatgpt on nlp tasks. *arXiv preprint arXiv:2306.09719* (2023).
- [46] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Roziere, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv*

- preprint arXiv:2302.13971* (2023).
- [47] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. LLaMA: Open and Efficient Foundation Language Models. *arXiv:2302.13971* [cs.CL] <https://arxiv.org/abs/2302.13971>
- [48] Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R Bowman. 2018. GLUE: A multi-task benchmark and analysis platform for natural language understanding. *arXiv preprint arXiv:1804.07461* (2018).
- [49] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. 2022. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682* (2022).
- [50] Zhengqi Xu, Ke Yuan, Huiqiong Wang, Yong Wang, Mingli Song, and Jie Song. 2024. Training-free pretrained model merging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5915–5925.
- [51] Prateek Yadav, Derek Tam, Leshem Choshen, Colin A Raffel, and Mohit Bansal. 2023. Ties-merging: Resolving interference when merging models. *Advances in Neural Information Processing Systems* 36 (2023), 7093–7115.
- [52] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai,
- Sinan Tan, Tianhang Zhu, Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zhihao Fan. 2024. Qwen2 Technical Report. *arXiv preprint arXiv:2407.10671* (2024).
- [53] Enneng Yang, Zhenyi Wang, Li Shen, Shiwei Liu, Guibing Guo, Xingwei Wang, and Dacheng Tao. 2023. Adamerging: Adaptive model merging for multi-task learning. *arXiv preprint arXiv:2310.02575* (2023).
- [54] Peng Ye, Chenyu Huang, Mingzhu Shen, Tao Chen, Yongqi Huang, Yuning Zhang, and Wanli Ouyang. 2023. Merging Vision Transformers from Different Tasks and Domains. *arXiv:2312.16240* [cs.CV] <https://arxiv.org/abs/2312.16240>
- [55] Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. 2024. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *Forty-first International Conference on Machine Learning*.
- [56] Jihai Zhang, Xiaoye Qu, Tong Zhu, and Yu Cheng. 2025. CLIP-MoE: Towards Building Mixture of Experts for CLIP with Diversified Multiplier Upcycling. *arXiv:2409.19291* [cs.CV] <https://arxiv.org/abs/2409.19291>
- [57] Tong Zhu, Daize Dong, Xiaoye Qu, Jiacheng Ruan, Wenliang Chen, and Yu Cheng. 2024. Dynamic Data Mixing Maximizes Instruction Tuning for Mixture-of-Experts. *arXiv:2406.11256* [cs.CL] <https://arxiv.org/abs/2406.11256>
- [58] Tong Zhu, Xiaoye Qu, Daize Dong, Jiacheng Ruan, Jingqi Tong, Conghui He, and Yu Cheng. 2024. LLaMA-MoE: Building Mixture-of-Experts from LLaMA with Continual Pre-training. *arXiv:2406.16554* [cs.CL] <https://arxiv.org/abs/2406.16554>